CLASS 6 — MATHS · FORMULA SHEET

Chapter 3: **Number Play** | Quick revision, one page

Making numbers from digits

- Largest number: arrange the digits in **descending** order.
- Smallest number: arrange in ascending order, but a non-zero digit must come first (no leading 0).
- Even number: last digit is 0, 2, 4, 6 or 8. Odd number: last digit is 1, 3, 5, 7 or 9.
- Successor $= n + 1$; predecessor $= n - 1$.

Digit games

- Palindrome: reads the same forwards and backwards, e.g. 121, 4334.
- Reverse-and-add: add a number to its reverse; repeat until a palindrome appears.
- Kaprekar step: (largest arrangement of digits) $-$ (smallest arrangement). For 4-digit numbers, repeating reaches 6174.
- Digit sum: add all the digits; a number is divisible by 3 (or 9) if its digit sum is.

Hailstone (Collatz) sequence

- If n is even, next term $= \frac{n}{2}$. If n is odd, next term $= 3n + 1$.
- Example: $6 \to 3 \to 10 \to 5 \to 16 \to 8 \to 4 \to 2 \to 1$ (8 steps).

Supercells

- Each cell equals the sum of the two cells just below it. Work upward from the bottom row; to find a missing cell, subtract.

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